

PL-7008: Create Agents with Microsoft Copilot Studio

Based on Microsoft Learn

<https://learn.microsoft.com/en-us/training/paths/create-extend-custom-copilots-microsoft-copilot-studio/>

In This Course...

- Learn how to develop agents using Microsoft Copilot Studio
- Understand agent components: tools, , topics, entities, triggers, and channels
- Hands-on labs using Microsoft Copilot Studio to create agents in a lab environment
- **Note: The Microsoft Copilot Studio UI is frequently updated. Lab instructions may not always match the current interface exactly. Please keep this in mind.**

Time Table (Planned)

Morning Session: 9:30am – 11:30am

- ✓ Opening & Lab Setups
- ✓ Lecture

Lunch Hour + Lab session: 11:30am – 2:00pm

- ✓ Lab Module 1, 2
- ✓ Lunch

Afternoon Session: 2:00 – 5:30pm

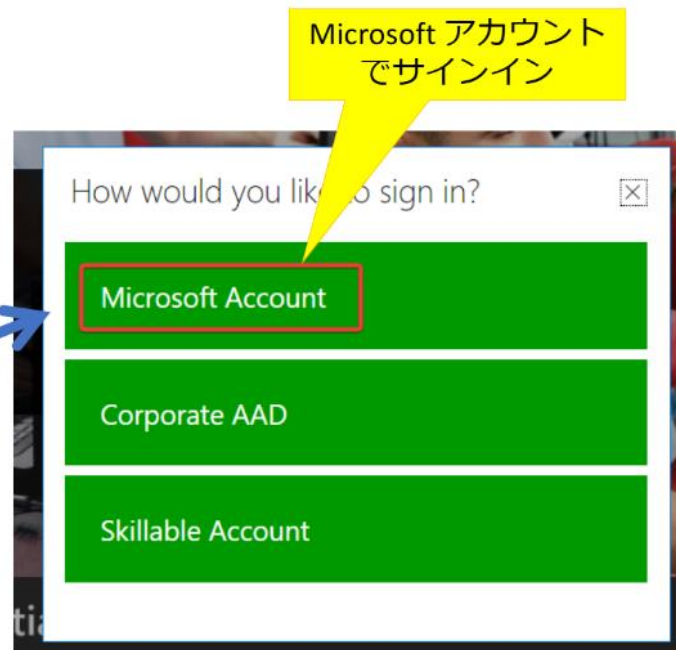
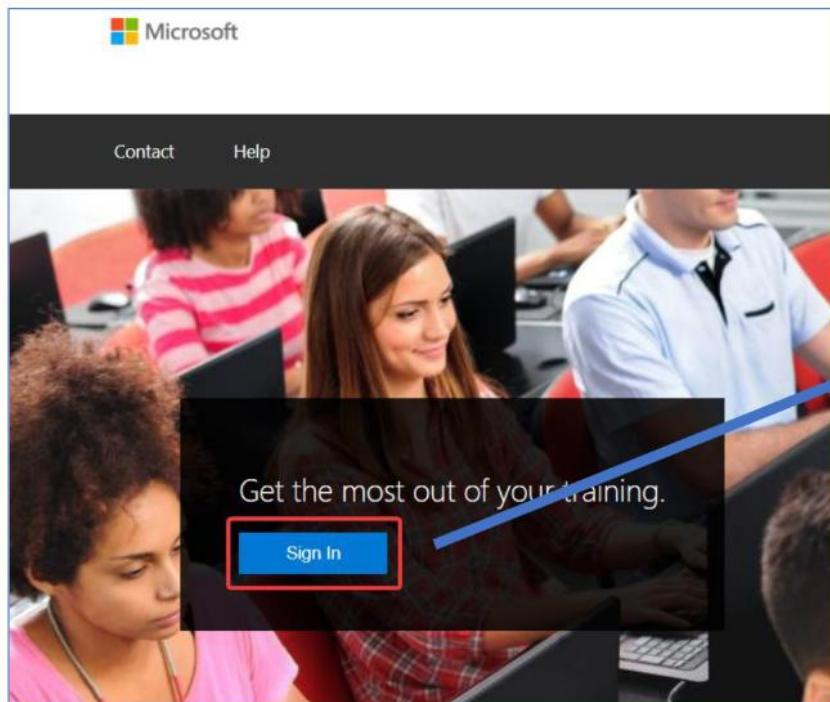
Lecture: 2:00-3:30pm

Summary: 3:30-4:00pm

Lab Module from 3 to whatever you can finish: 4:00 – 5:30pm

Lab Setups

URL : <https://esi.learnondemand.net>



Lab Setups

 Microsoft

Ayaka ▾ 

Admin ▾ My Dashboard My Calendar Contact Help

 Current Training
Ayaka Yamazaki

 [Details](#)  [Edit](#)

 [Transcript](#)  [Redeem Training Key](#)

All times shown in Line Islands Standard Time

Lab Setups

← ↻ <https://esi.learnondemand.net/TrainingKey/Redeem>

 Microsoft

Admin ▾ My Dashboard My Calendar Contact Help

 Redeem Training Key

Training Key: *

[Redeem Training Key](#)

Lab Setups

← ↻ https://esi.learnondemand.net/ClassEnrollment/5812470



Admin ▾

My Dashboard

My Calendar

Contact

Help



Enrollment

Ayaka Yamazaki

(ID 99957) **PL-7008**: Create and extend custom copilots in Microsoft Copilot Studio [Cloud Slice Provided] - ESI

Basic Information

Student: [Ayaka Yamazaki](#) Details ▾

Event: [\(ID 99957\) PL-7008: Create and extend custom copilots in Microsoft Copilot Studio \[Cloud Slice Provided\] - ESI](#) Details ▾

Enrollment Status: **Enrolled**

Completion Status: **Unknown**

Expires: **Saturday, November 14, 2026**

Is Retake: **Yes**

Enable Labs: **Yes**

Activities

Lab Structure

- **Labs 1 & 2: Create a Power Platform environment and build an Expense Claim Support Agent**
 - Relatively straightforward — hands-on lab recommended
- **Labs 3–10: Build a Real Estate Booking Agent**
 - More advanced — approximately 5 hours total
 - Labs 3–10 must be completed sequentially

Lab



Admin ▾

My Dashboard

My Calendar

Contact

Help



Create agents in Microsoft Copilot Studio

All Learning Paths

Duration:

8 Hours

Lab Series:

PL-7008: Create agents in Microsoft Copilot Studio

Virtualization Platform: Hyper-V

RAM:

8.5GB

Content Version:

2

Is Exam:

No

Status:

Not Running

Launch



Create agents in Microsoft Copilot Studio

All Learning Paths

Duration:

8 Hours

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No

Status:

Not Running

Launch

Power Platform admin center Home - Microsoft Copilot Studio

https://copilotstudio.microsoft.com/environments/0bfeaa17-1a9a-e23b-8eb4-99c954d1b10b/home

Copilot Studio

Environment Ayaka Copilot (Ctrl+Shift+J)

What would you like to build?

Agent Workflow

Start building by describing what your agent needs to do

Start building from scratch



Agent

Flexible solutions that take action, share knowledge, and handle tasks



Computer-using agent

Let agents accomplish even more across apps and websites



Create workflow

Powerful automations speed up your processes with ease and security in mind

Create agents in Microsoft Copilot Studio

Exit Lab

Save Progress And Exit

Instructions Res...

option to s

End Lab

2. Close any ... displayed.
3. Select **+New** to create a new environment with the following settings:
 - Type: Developer
 - Region: **default region**
 - Name: *Your name*
4. Select **Next**
5. Populate the following settings:
 - Language: English (United States)
 - Currency: USD (\$)
 - Deploy sample apps and data: No
6. Select **Save** and wait until the state of your environment is **Ready** (you can use the **Refresh** button to update the display)
7. Select the newly created environment
8. In the **Details** section, copy the **Environment ID** value.
9. In a new browser tab, enter `https://copilotstudio.microsoft.com/environments/<Environment ID>`, replacing `<Environment ID>` with the value you copied. **Note:** Save this URL or add it to your Favorites Bar so you can use it later to navigate directly to your environment. There is currently an issue opening Copilot Studio in the default environment.
10. If prompted, sign in with your credentials and skip any welcome messages.
11. **Skip to Task 2 in the next lab to build your first agent.**

Previous Next

7 Hr 53 Min Remaining

Would you like to save your lab and close it for now?

Your lab will be saved for 7 days. It will expire on 5/24/2026, 5:12:27 PM. You can save up to 5 labs.

Yes, save my lab

No, not yet

Your lab is saved

Your lab has been saved and will be available for you to resume for the next 7 days. It will expire 5/24/2026, 5:13:39 PM.

Close Window

Create agents in Microsoft Copilot Studio

Exit Lab

Instructions Resources

Create an environment

⚠ Currently, **Power Platform** Environments are experiencing undesired delays and errors when being provisioned. While this issue is under investigation by **Microsoft**, note that you may encounter the error **An unexpected error occurred** after logging in. Please follow **Step 3** below to create the environment, and if you do not see the environment created after several minutes, proceed to **Step 5**. You should see a message indicating your environment is being connected. Before you start the lab exercise, let's create a development environment for you to work in.

1. Open a web browser, navigate to `https://admin.powerplatform.microsoft.com`, and sign in as `admin@MMLCPx65963017.onmicrosoft.com` using `z;-b25-q1[K~#3MU6]0201OxXW)SAtf` as the password. If prompted, choose the option to stay signed in.
2. Close any pop-up messages that are displayed.
3. Select **+New** to create a new environment with the following settings:
 - Type: Developer
 - Region: *default region*
 - Name: *Your name*

← Previous

Next →

7 Hr 51 Min Remaining

Admin

Student

ENG

Admin

Password →

PL-7008 Table of Contents

Module Name	Labs
Getting Started with Microsoft Copilot Studio	1, 2
Building Effective Agents with Microsoft Copilot Studio	—
Enhancing Microsoft Copilot Studio Agents	—
Building Autonomous Agents in Microsoft Copilot Studio	—
Building Your First Agent with Microsoft Copilot Studio	3, 4
Managing Topics in Microsoft Copilot Studio	5, 6
Using Entities and Variables in Microsoft Copilot Studio	7
Creating Agents with Microsoft Copilot Studio & Dataverse for Teams	8, 9, 10

MODULE 1

Getting Started with Microsoft Copilot Studio

Module 1 — Topics

- What is an Agent?
- What is Microsoft Copilot Studio?
- Three Types of Microsoft Copilot Studio
- What is Power Platform?
- What is a Power Platform "Environment"?
- Summary

What Is an Agent?

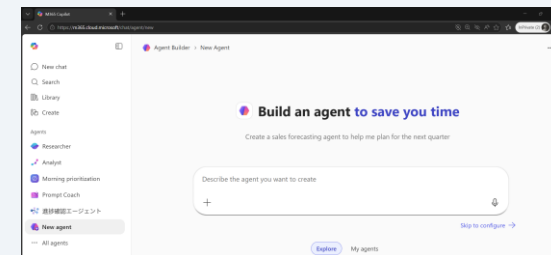
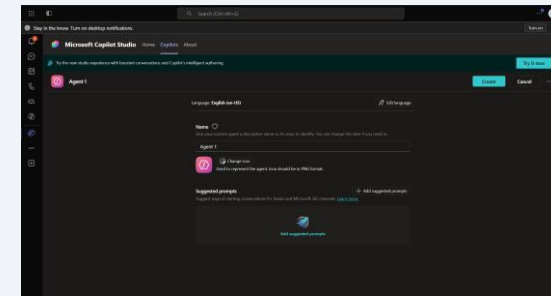
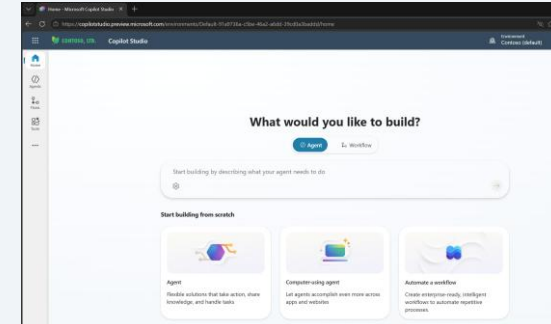
- **"Agent" = a representative or intermediary**
- Uses generative AI
- Can converse with humans through a chat interface
- Not just answering questions — agents can use tools to perform tasks on behalf of the user:
 - Code Interpreter — generate and run Python code for data analysis
 - Image Generator — create images
 - Power Automate Flows — send emails on behalf of users, etc.

What Is Microsoft Copilot Studio?

- A development tool for building agents
- Access at: <https://copilotstudio.microsoft.com/>
- Supports no-code / low-code development
 - Uses a graphical UI — no Python or JavaScript programming required
 - Relatively easy to develop agents

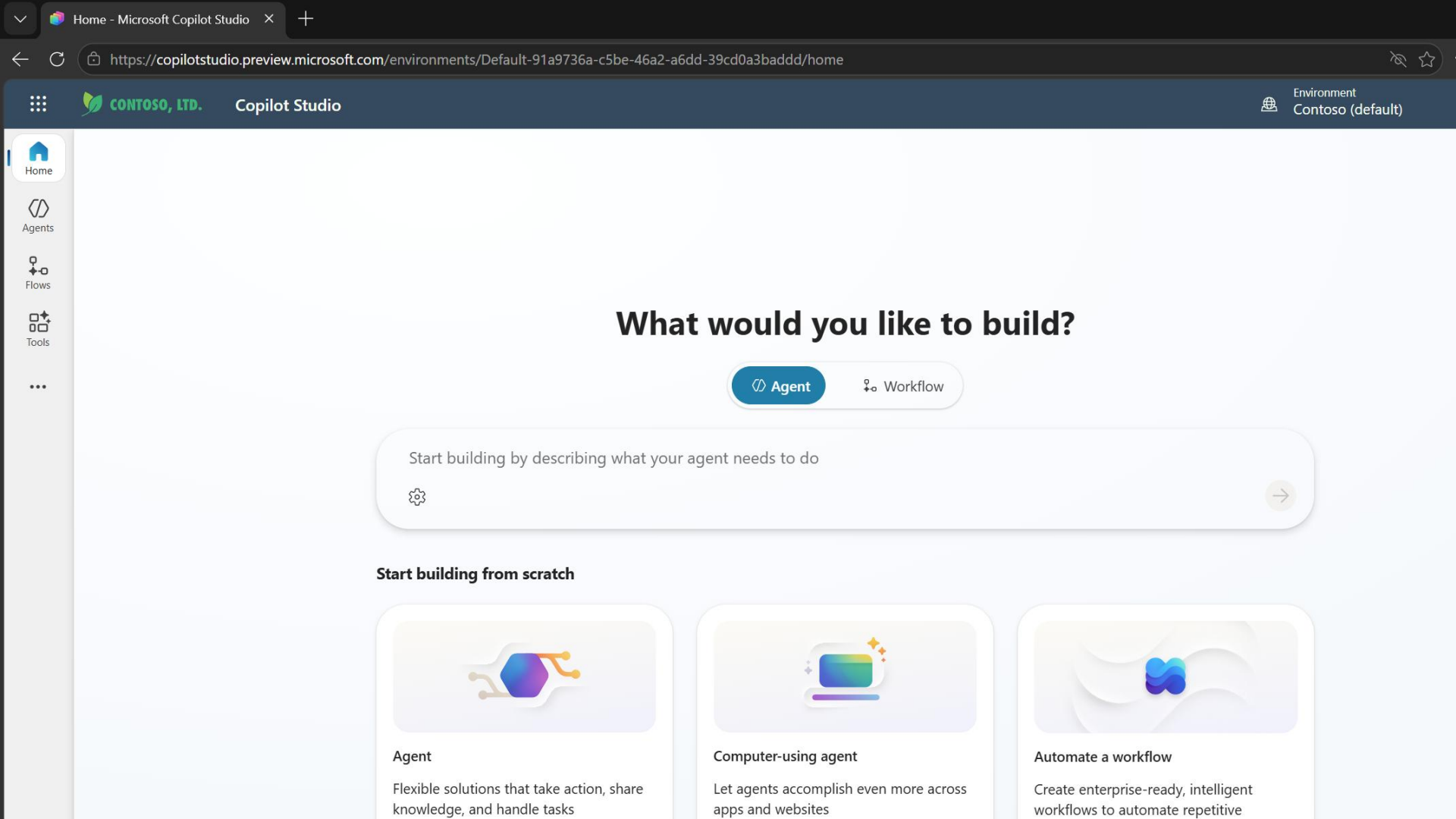
Three Types of Microsoft Copilot Studio

- **1. Microsoft Copilot Studio (Web App) — "Full Experience"**
 - Access via <https://copilotstudio.microsoft.com/>
 - Create full-featured agents — this is the focus of this course
- **2. Microsoft Copilot Studio (Teams App)**
 - Added as a Teams app — can only create "classic chatbots"
 - Not covered in this course
- **3. Microsoft Copilot Studio "Lite Experience" (Agent Builder)**
 - Launch from "Create an agent" link in Microsoft 365 Copilot Chat
 - Create simplified agents — covered in MS-4019 course



Comparison of the Three Types

Feature	Microsoft Copilot Studio Web App (Full)	Microsoft Copilot Studio Teams App	Agent Builder
Required license	Copilot Studio(Credits/PAYG)	Microsoft 365(Dataverse for Teams)	Microsoft 365 Copilot
Channels	Copilot Chat, Teams, Web App, Facebook, Slack, LINE, etc.	Limited for Dataverse for Teams	Teams, Copilot Chat
-based answers (Word, Website, etc)	Yes	No	Yes
Topics (process flows)	Yes	Yes	No
How to chose topics	Generative AI Orchestration <i>Uses generative AI to select topics and generate responses based on connected sources</i>	Classic Orchestration <i>Selects a topic by matching the user's input to predefined trigger phrases that are closest to the utterance.</i>	M365 Copilot Managed
Storage	Dataverse	Dataverse for Teams	M365-managed storage



- Home
- Agents
- Flows
- Tools
- ...

What would you like to build?

Agent

Workflow

Start building by describing what your agent needs to do



Start building from scratch



Agent

Flexible solutions that take action, share knowledge, and handle tasks



Computer-using agent

Let agents accomplish even more across apps and websites

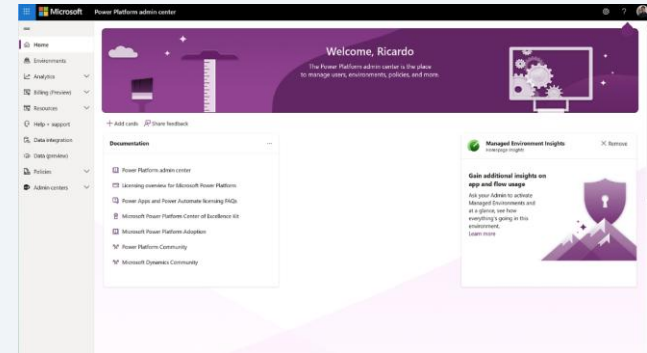


Automate a workflow

Create enterprise-ready, intelligent workflows to automate repetitive

What Is Power Platform?

- A low-code / no-code development platform from Microsoft
- Build apps, automate processes, analyze data, and create agents without specialized programming
 - Power BI — Data visualization & analytics
 - Power Apps — Business application development
 - Power Automate — Business process automation
 - Power Pages — Website creation & publishing
 - **Microsoft Copilot Studio — Agent development (formerly Power Virtual Agents)**



What Is a Power Platform "Environment"?

- A container for managing Power Platform apps, data, and agents
- Multiple environments can be created (e.g., Development, Production)
- A geographic location is specified during creation — data is stored in that region's data center
- Environments are managed under an Entra ID tenant
- Create and manage environments via the Power Platform Admin Center
 - <https://admin.powerplatform.com/>

Tenant, Environment, Solution, and Agent

- **Entra ID Tenant (manages user accounts and environments for the organization)**
 - **Power Platform Environment (Development, Production, etc.)**
 - Lab Exercise: Common Data Service Default Solution
→ Expense Claim Support Agent
 - Lab Exercise : Bookings
→ Real Estate Booking Agent
 - ✓ Dataverse Tables (Real Estate Property, Booking Request)
 - ✓ sources

Tenant, Environment, Solution, Agent

Entra ID Tenant (= where user accounts and environments of an organization are managed)

Power Platform Environment (Development, Production, etc.)

Common Data Services Default Solution

Solution

Expense Request Support Agent



Knowledge

Bookings

Solution

Dataverse Table

Real Estate Property

Dataverse Table

Booking Request

Real Estate Viewing Reservation Agent

MODULE 5

Building Effective Agents with Microsoft Copilot Studio

Key Use Cases for Agents

- **Agents that support employees – “Employee-facing”**
 - e.g., An agent to help onboard new employees
 - Saves employee time
- **Agents that support customers – “Customer-facing”**
 - e.g., An agent on a website to handle customer inquiries
 - Automates first-level support so humans can focus on complex issues

Note: The choice between employee-facing and customer-facing impacts design decisions (authentication, channels, tone, etc.).

Agent Handoff (Escalation)

- When an agent cannot resolve a user's issue, it can hand off to a human operator
- This is called "**handoff**" or "escalation"
- Microsoft Copilot Studio agents support this feature out-of-the-box (disabled by default)
 - 💡 Nice to consider especially for customer-facing agents to ensure issues are resolved to maintain customer satisfaction.

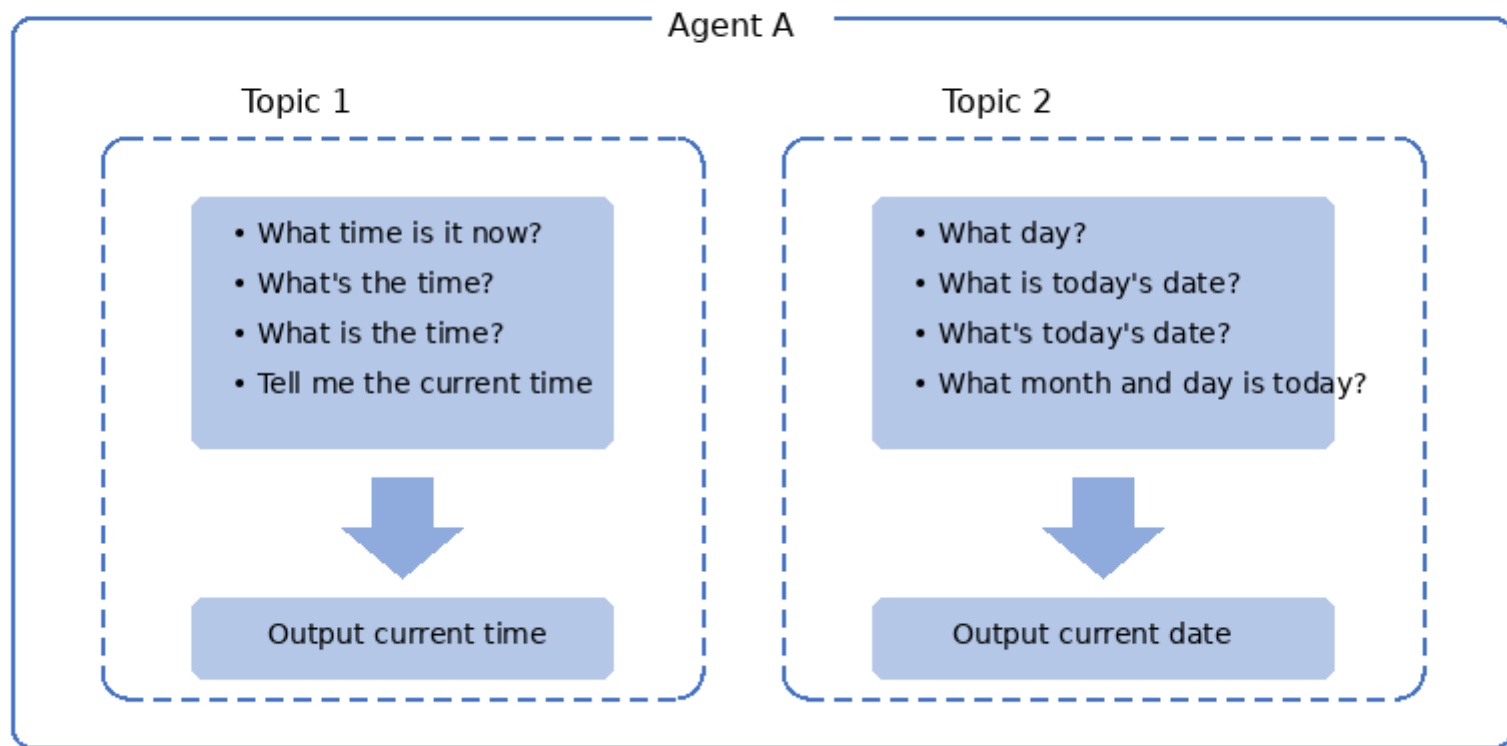
Agent Design Considerations

- **Evaluate the impact of the agent**
 - Focus on automating tasks that currently cost users significant time/effort
- **Choose the right channels**
 - Consider where users will access the agent: Teams, Web, Slack, LINE, etc.
- **Identify the required data ()**
 - Attach files or connect external data sources to the agent

Topics: What an Agent Can Do

- A topic represents a specific task or conversation the agent can handle
- Each topic is triggered by user input matching certain phrases
- **Example:**
 - Topic 1 — "What time is it?" → Output the current time
 - Topic 2 — "What day is it?" → Output the current date

Topic: Tasks an Agent Can Perform



System Topics vs. Custom Topics

System Topics

Built-in Topics

- Automatically provided by the system
- Handle common conversational needs
- Examples: Greeting, Fallback, Escalation
- You usually don't design these from scratch

**You can customize, but they already exist.*

Note:

Think of System Topics as the operating system

Custom Topics

Your Topics

- Created by you for business scenarios
- Represent specific tasks or processes
- Examples: Check asset status, Submit a request
- **This is where your business logic lives**

Note:

Think of System Topics as your business applications.

Topics - V2_Expense Claim Assis... X

← → 🔍 https://copilotstudio.preview.microsoft.com/environments/Default-91a7738a-c58e-46a2-af6d-39cda3badd5b0a/e4051451-af0d-1111-bac5-70ada59bde4b/tools/system-topics

Environment: Contoso (default)

V2_Expense Claim Assistant Overview Knowledge Tools Agents Topics Activity Evaluation Analytics +1 Published 5/16/2026 Publish Settings Test

+ Add a topic 🔍 Search system topics

All Custom (5) System (3) Last refreshed 19 minutes ago

System topics are necessary to help your agent run effectively. Without them, your agent may not work as intended. You can't delete system topics.

Name	Type	Trigger	Last modified	Editing	Errors	Blocked	Enabled
Escalate	System Topic	On Talk to Representative	MOD Administrator 20 hou...				On
When an error occurs	System Topic	On error	MOD Administrator 20 hou...				On
Sign in	System Topic	On Sign In	MOD Administrator 20 hou...				On
Fallback	System Topic	Unknown topic	MOD Administrator 20 hou...				On
Enhance conversations	System Topic	Unknown topic	MOD Administrator 20 hou...				On

Topics - V2_Expense Claim Assis... X

← → 🔍 https://copilotstudio.preview.microsoft.com/environments/Default-91a7738a-c58e-46a2-af6d-39cda3badd5b0a/e4051451-af0d-1111-bac5-70ada59bde4b/tools/custom-topics

Environment: Contoso (default)

V2_Expense Claim Assistant Overview Knowledge Tools Agents Topics Activity Evaluation Analytics +1 Published 5/16/2026 Publish Settings Test

+ Add a topic 🔍 Search custom topics

All Custom (5) System (3) Last refreshed 18 minutes ago

Name	Type	Trigger	Last modified	Editing	Errors	Blocked	Enabled
Ask about expenses contact	Topic	By agent	MOD Administrator 20 hou...				On
Greetings	Topic	By agent	MOD Administrator 20 hou...				On
Thank you	Topic	By agent	MOD Administrator 20 hou...				On
Thank you for your inquiry	Topic	By agent	MOD Administrator 20 hou...				On
Start over	Topic	By agent	MOD Administrator 20 hou...				On

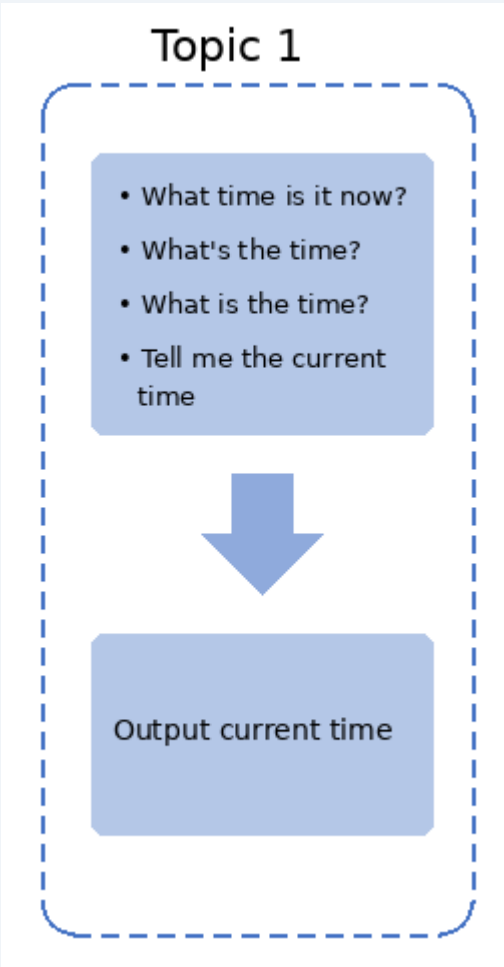
Topic Design & Operations Considerations

- Identify the types (scenarios) of topics — e.g., information retrieval, task execution, troubleshooting
- Design topics based on how users will actually interact with the agent
- Repeatedly test and validate that topics work correctly
- Monitor usage in production and continuously improve

Trigger Phrase Considerations

- **Trigger phrases activate/select topics based on user input**
- Set appropriate trigger phrases for each topic
- Anticipate how users will ask questions and register diverse phrase patterns
- Note: When Generative AI Orchestration is enabled, trigger phrases are not used. Instead, set appropriate topic names and descriptions — the AI will select the best-matching topic.

Topic 1

- 
- ```
graph TD; A["• What time is it now?
• What's the time?
• What is the time?
• Tell me the current time"] --> B["Output current time"]
```
- What time is it now?
  - What's the time?
  - What is the time?
  - Tell me the current time

Output current time

MODULE 6

# Enhancing Microsoft Copilot Studio Agents

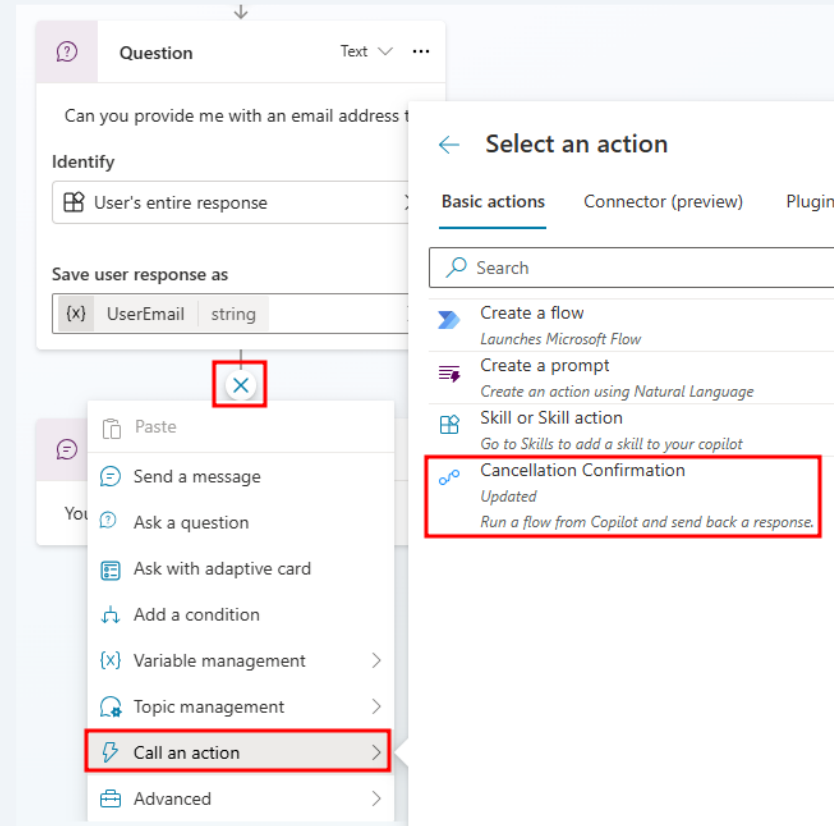
# Three Ways to Enhance Agent Capabilities

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- **1. Create actions that launch Power Automate flows**
  - Automate operations like sending emails via Outlook
  - Add a "Call an action" node within a topic to trigger a flow
- **2. Hand off conversation to a live chat operator**
  - When the agent can't resolve an issue, transfer to a human operator
  - Conversation context is passed to the customer engagement app
- **3. Use generative answers**
  - Agent gathers information from various sources and generates answers using AI
  - Uses RAG (Retrieval-Augmented Generation)

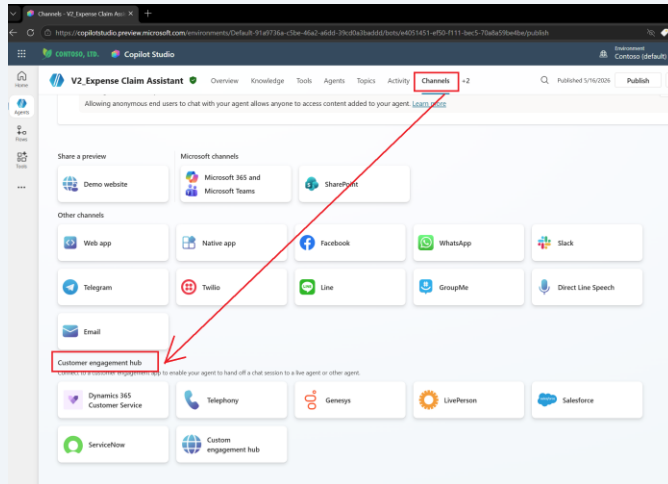
# Enhancement (1): Power Automate Flow Actions

- **Power Automate flows** can automate various operations
  - Example: "Send an email using Outlook"
- Add a "Call an action" node within an agent topic
- Select an existing Power Automate flow or create a new one
- When this topic executes, the flow runs (e.g., an email is sent)
- Required data (recipient, subject, etc.) is collected by the agent asking the user during the conversation



# Enhancement (2): Handoff to Live Chat Operator

- When an agent cannot resolve a customer issue during first-level support, it can hand off to a human operator
- **The agent passes the first-level support conversation data to the human operator**
- Implemented by passing data to a **customer engagement app** used by the operator
- This allows the human operator to continue the support with full context



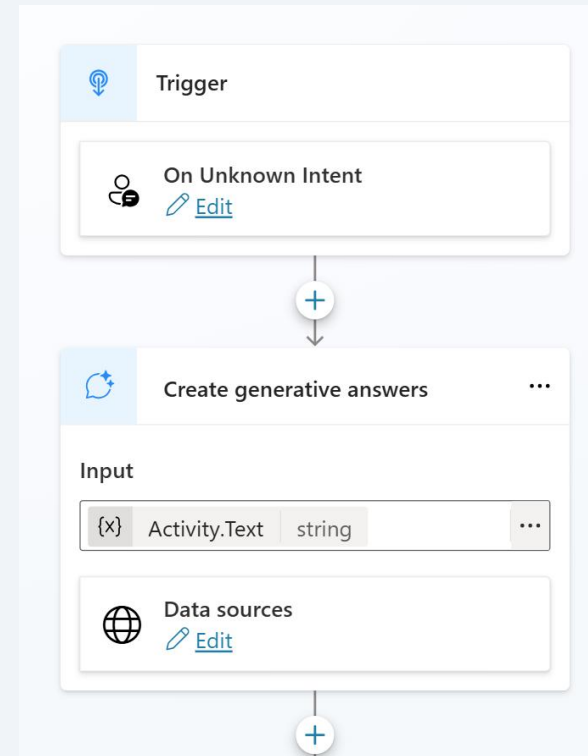
(i.e) [Configure handoff to Dynamics 365 Customer Service](#)



# Enhancement (3): Generative Answers

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- Agent gathers information from various sources and uses generative AI to create answers
- Use the "Create a generative answer" node within a topic
- **Easily implement RAG (Retrieval-Augmented Generation)**
- Supported sources:
  - Public websites, Dataverse, Files, SharePoint, OneDrive
- Note: With Generative AI Orchestration enabled, the agent uses directly without needing this node in a topic



# Add knowledge



Help your agent provide more relevant information and insights. [Learn more](#)

🔍 Search for a knowledge source



⬆ Upload file

Drag and drop, or [select to browse](#), or upload and sync from



OneDrive



SharePoint

Helps your agent access knowledge from remote file locations. [Learn more](#)

★ Featured

🔗 Advanced



See suggestions



Public websites



SharePoint

Powered by Work IQ



Azure AI Search



Dataverse



Dynamics 365



Salesforce

now.

ServiceNow



Azure SQL

MODULE 8

# Building Autonomous Agents in Microsoft Copilot Studio

# Autonomous Agents (October 2024+)

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- In October 2024, Microsoft announced the ability to create autonomous agents in Copilot Studio
- More customers can now use AI to reengineer critical business processes
- Microsoft also started offering pre-built autonomous agents

# What Are Autonomous Agents?

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- **Traditional agents - start processing when a user sends a prompt**
  - Without a prompt, the agent doesn't do anything
- **Autonomous agents - can run business processes without waiting for user input**

Example:

- An agent that acts when a new email arrives in an Outlook inbox
- This agent can handle email responses without any human instruction

# Examples of Autonomous Agent Triggers

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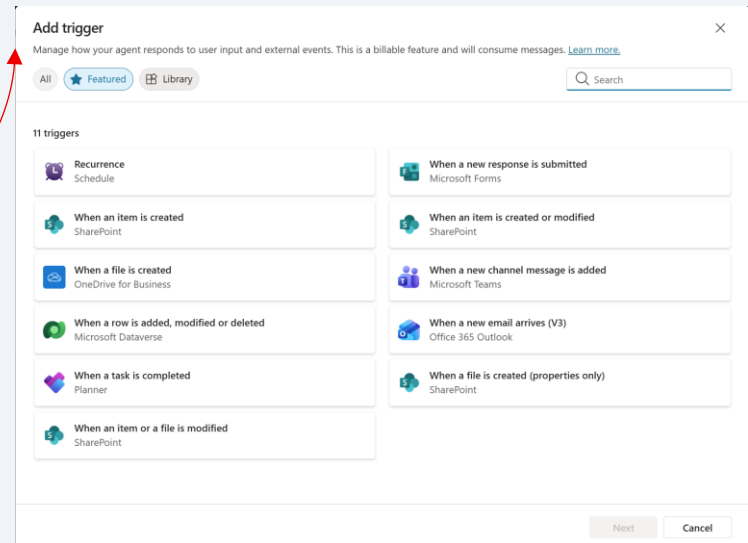
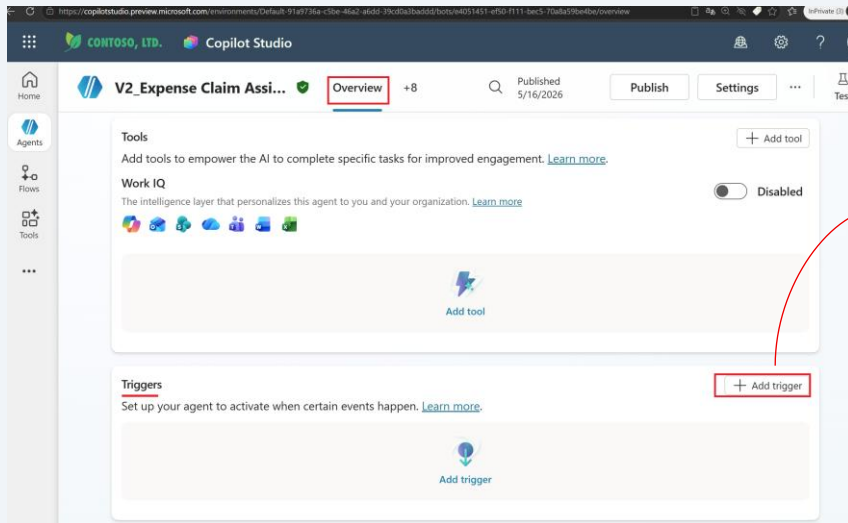
- When a new email arrives in an Outlook inbox
- When an item is created in SharePoint
- When a file is created in OneDrive
- When a task is completed in Planner
- When a new row is added to a Dataverse table
- When a specified time interval elapses

Note:

*These triggers are provided by Power Automate connectors, not by the Autonomous Agent itself.  
As of February 2026, 845+ trigger types are available across Power Automate connectors*

# Summary: Autonomous Agents

- Autonomous agents can be created to automatically respond to specific events
  - Example: "When a new email arrives in the Outlook inbox, do \_\_\_\_"
- These agents operate without human direction, handling tasks proactively



**Labs 1 & 2**

**Instructor Demonstration**



## Power Platform Environment for Practice

Default Solution (Common Data Service Default Solution)



Expense Request Support Agent

Knowledge



Expense Request Policy.docx

Topic "Expense Inquiry Contact"

Trigger

"Who do I contact for  
expense inquiries?"



Send Message

Please email  
finance@contoso.com



# Labs 1 & 2: Overview

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- **Lab 1: Lab Setup**
  - Create a Power Platform environment
  - Access Microsoft Copilot Studio
- **Lab 2: Create an Agent Using Copilot Studio**
  - Create an "Expense Claim Support Agent"
  - Add a topic so the agent can answer expense inquiry questions
  - Add (a Word file with expense policies)
  - Publish the agent
  - Test the agent on a demo website

# System Topics: Conversational Boosting & Fallback

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- **"Conversational Boosting" topic**
  - Triggered when the agent can't identify an appropriate topic (unclear user intent)
  - Agent uses generative AI to attempt creating an answer from sources
  - If this fails, the "Fallback" topic is invoked
- **"Fallback" topic**
  - Outputs: "I'm sorry, I can't understand your inquiry. Please try rephrasing."
  - After 3 invocations, triggers the "Escalate" topic (handoff to a human)
- Conversational Boosting has higher priority than Fallback (both share the same trigger)

MODULE 2

# Building Your First Agent with Microsoft Copilot Studio

# What Is Dataverse?

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- The data foundation for Power Platform (Power BI, Power Automate, Copilot Studio, etc.)
- A cloud-based database for **recording various business data**
- Agents can access Dataverse to retrieve and store business data
- Formerly called "Common Data Service (CDS)"

# What Is a Dataverse Solution?

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- A ZIP file that bundles apps, flows, tables, security roles, and other components
- Used for migrating between Power Platform environments
  - Example: Export a solution from a test environment and import it into production
- Solutions can be exported and imported between environments

# Labs 3 & 4: Overview

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- **Lab 3: Import a Dataverse Solution**
  - Import the "Bookings" solution into the Power Platform environment
  - Register 2 sample real estate property records
- **Lab 4: Build the Real Estate Booking Agent**
  - Create the agent within the Bookings solution
  - Configure name, description, and instructions
  - Add a source (website with real estate sales tips)
  - Test: The "Conversational Boosting" topic runs and generates answers from the website

# </> Real Estate Viewing Reservation Agent

## Knowledge

### Dataverse Table



Real Estate Info  
(Inside "Bookings"  
Dataverse Solution)

### Entity

Property Type  
("House",  
"Apartment",  
"Condo")

## Topic "Customer Details"

Trigger

"Customer Info"

Question

Please enter your  
name. -> [Name]

Question

Please enter your  
email address.  
-> [Email]

Question

You entered [Name],  
[Email]. Is this  
correct?  
(Yes/No) -> [Confirm]

## Topic "Real Estate Viewing Reservation"

Trigger

"I want to reserve a condo viewing"

Go to Topic

Call Topic "Customer Details"

Condition

[Confirm]="Yes"

No

Yes

Question

Which property would you like to view? -> [Property]

Question

What date and time? -> [Date/Time]

Send Message

Your viewing reservation has been received



# Generative AI Orchestration Setting

---

- **When Enabled (default):**
  - Agent combines lookup, built-in AI , and topic execution for the best answer
  - If is available, agent uses it directly (topics may not execute)
- **When Disabled:**
  - Agent triggers the "Conversational Boosting" topic, which uses the "Create generative answer" node
- **Key point: With provided, both enabled and disabled settings produce -based answers — just via different mechanisms**

# Power Apps Model-Driven Apps

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- Business applications auto-generated from data model definitions
- By creating a data model in Dataverse, forms, views, and dashboards are built almost automatically
- Used in Lab 3 to register sample real estate property data into Dataverse tables

MODULE 3

# Managing Topics in Microsoft Copilot Studio

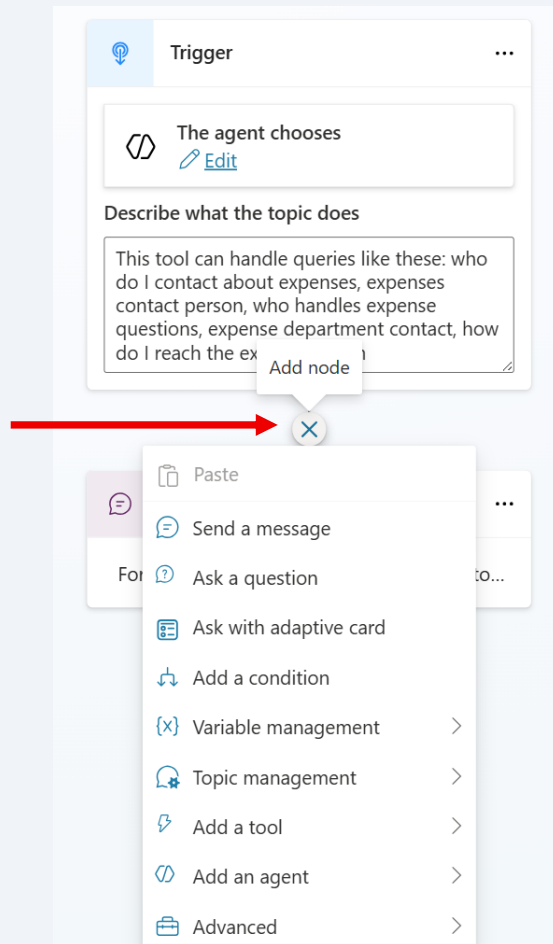
# About Topics - recall

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- **A topic = a specific task the agent can perform**
- Topics can range from simple to very complex (including conditional branches)
- Each topic consists of:
  - One trigger (how the topic is activated)
  - One or more nodes (processing units like messages, questions, conditions)

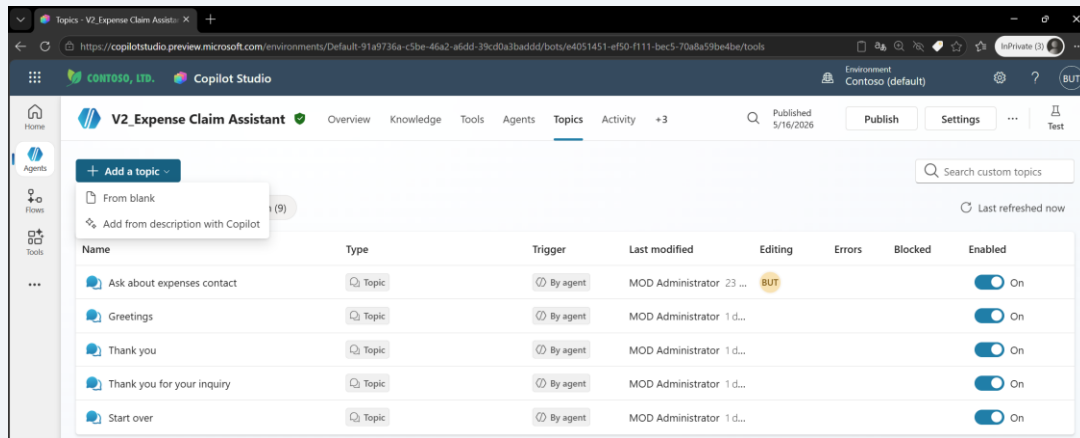
# What Is a Node?

- A node is a **processing steps** within a topic
  - they execute in sequence from top to bottom
- **Common node types:**
  - Message — Send a message to the user
  - Question — Ask the user for input
  - Condition — Branch based on variable values (IF/ELSE)
  - Action — Call a Power Automate flow or other action
  - Redirect — Navigate to another topic
  - Generative Answer — Generate an answer from sources



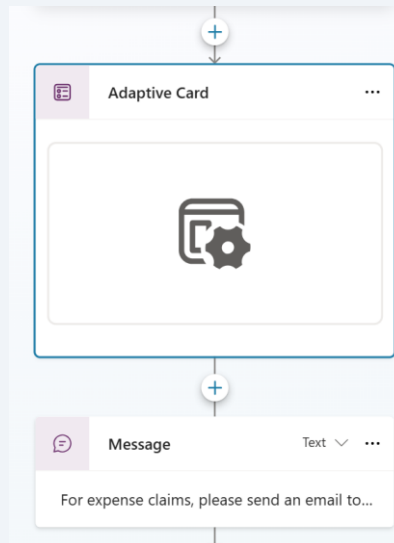
# Creating Topics with Copilot

- Microsoft Copilot Studio has a built-in Copilot (AI assistant)
- Enter prompts to automatically generate topics and nodes:
  - "Ask for name and email address"
  - "Add a thank-you message using the customer's name"
  - "Display the entered information using an Adaptive Card"
- **Very convenient — no manual node configuration required for common patterns**



# What Are Adaptive Cards?

- Adaptive Cards are a cross-platform card framework developed by Microsoft.
- They render natively in Teams, Outlook, Power Automate, and Copilot Studio.
- Adaptive Cards can also include interactive elements like buttons and input fields, making them useful for collecting structured data.
- Defined using JSON



# Labs 5 & 6: Overview

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- **Creating a "Customer Details" topic (using Copilot AI)**
  - Asks for customer name and email address
  - Displays an Adaptive Card to confirm the entered information
- **Creating a "Book a Real Estate Showing" topic (manually)**
  - Sets trigger phrases for booking requests
  - Calls the Customer Details topic to collect user info
  - Adds conditional branching (Yes/No confirmation)
  - Asks for property name and preferred date/time



# Topic Variables and Cross-Topic Data Passing

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- Variables within a topic can be configured to pass data to other topics
- **Example variables in the Customer Details topic:**
  - Name — Customer's name
  - EmailAddress — Customer's email
  - DetailsCorrect — Confirmation (Yes/No)
- When the Real Estate Booking topic calls the Customer Details topic, these variable values are passed back automatically

# Real Estate Booking Topic: Flow

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- 1. User says "I'd like to book a property showing"
- 2. Agent greets user and explains assistance
- 3. Redirects to Customer Details topic → collects name, email, confirmation
- 4. Condition: If DetailsCorrect = Yes:
  - Ask "Which property do you want to see?" → store in PropertyName variable
  - Ask "What date and time?" → store in DateTime variable
  - Display "Your showing has been scheduled!"
- 5. Publish and test on demo website

MODULE 4

# Using Entities and Variables in Microsoft Copilot Studio

# What Are Entities?

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- Entities represent types of data that the agent can automatically recognize in user text.
- Pre-built entities include: dates, times, numbers, email addresses, phone numbers, URLs, person names, and more.

<Example>

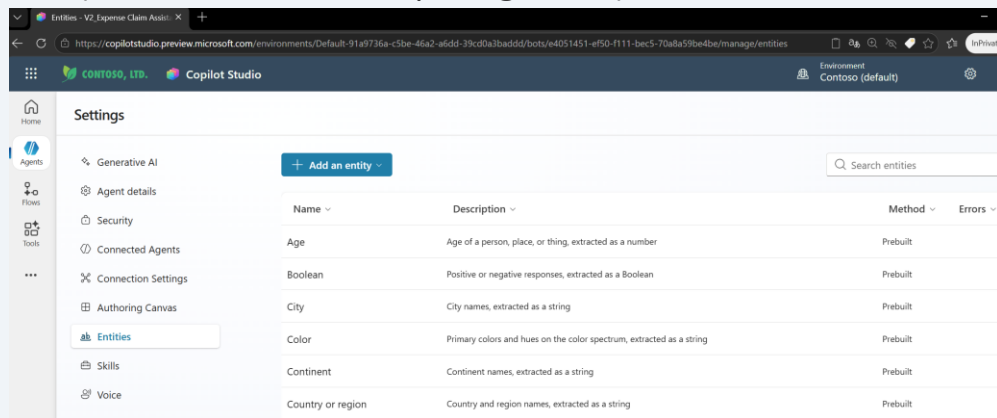
"My name is Ayaka. Please contact [ayaka@email.com](mailto:ayaka@email.com)"

→ Agent recognizes "Ayaka" as a person's name and the email as an email address

# Adding Custom Entities

- You can create new custom entities for your business needs
- Example: "Property Type" entity with values:**
  - Apartment (**synonyms: Flat**)
  - Condominium (synonyms: Townhouse)
  - House (synonyms: Single-family home)
- Smart Matching: When enabled, the agent uses fuzzy logic to interpret user input**
  - e.g., "Appartment" is interpreted as "Apartment" (auto-corrects minor spelling errors)

**Synonyms** allow you to map different expressions with the same meaning to a single entity value. This ensures that the agent can correctly interpret user input even when users use varied wording or alternative terms.



# What Are Variables?

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- Temporary "containers" that store recognized entities and other data within a topic
- Can be used in subsequent processing steps:
  - Composing messages (e.g., "Thank you, {Name}!")
  - Saving data to Dataverse
  - Passing data to other topics or flows

## Example:

"My name is Ayaka, my email address is Ayaka@email.com"

→ Stored in UserName and userEmail variables

*Variable flow:*

*user input*

*>> entity recognition*

*>> variable storage*

*>> subsequent use in messages or logic.*

# Lab 7: Working with Entities

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- View pre-built entities in agent Settings → Entities
- Create a "Property Type" entity (Closed list):
  - Values: Apartment, Condominium, House
  - Add synonyms for each value
  - Enable Smart Matching for fuzzy input interpretation
- Create a "Number of Bedrooms" entity using Regular Expressions (Regex)

MODULE 7

# Creating Agents with Microsoft Copilot Studio & Dataverse for Teams



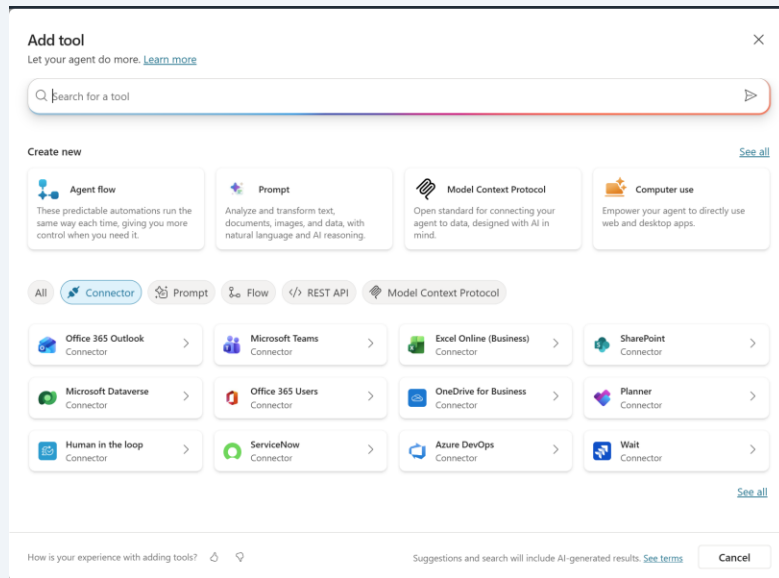
# Microsoft Dataverse for Teams

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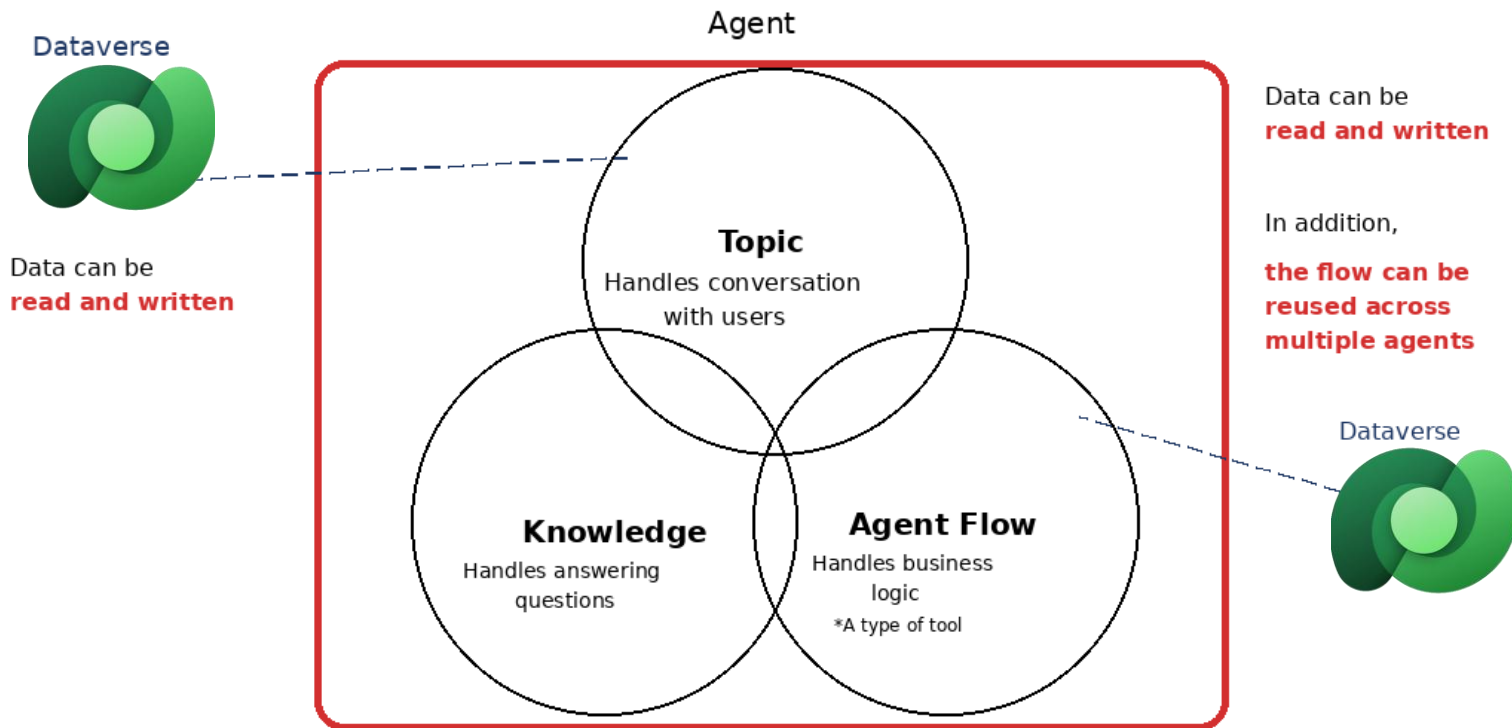
- A simplified version of Dataverse available within Teams
- Included with any Microsoft 365 license
- Agents can access Dataverse for Teams data

# What Are Agent Flows?

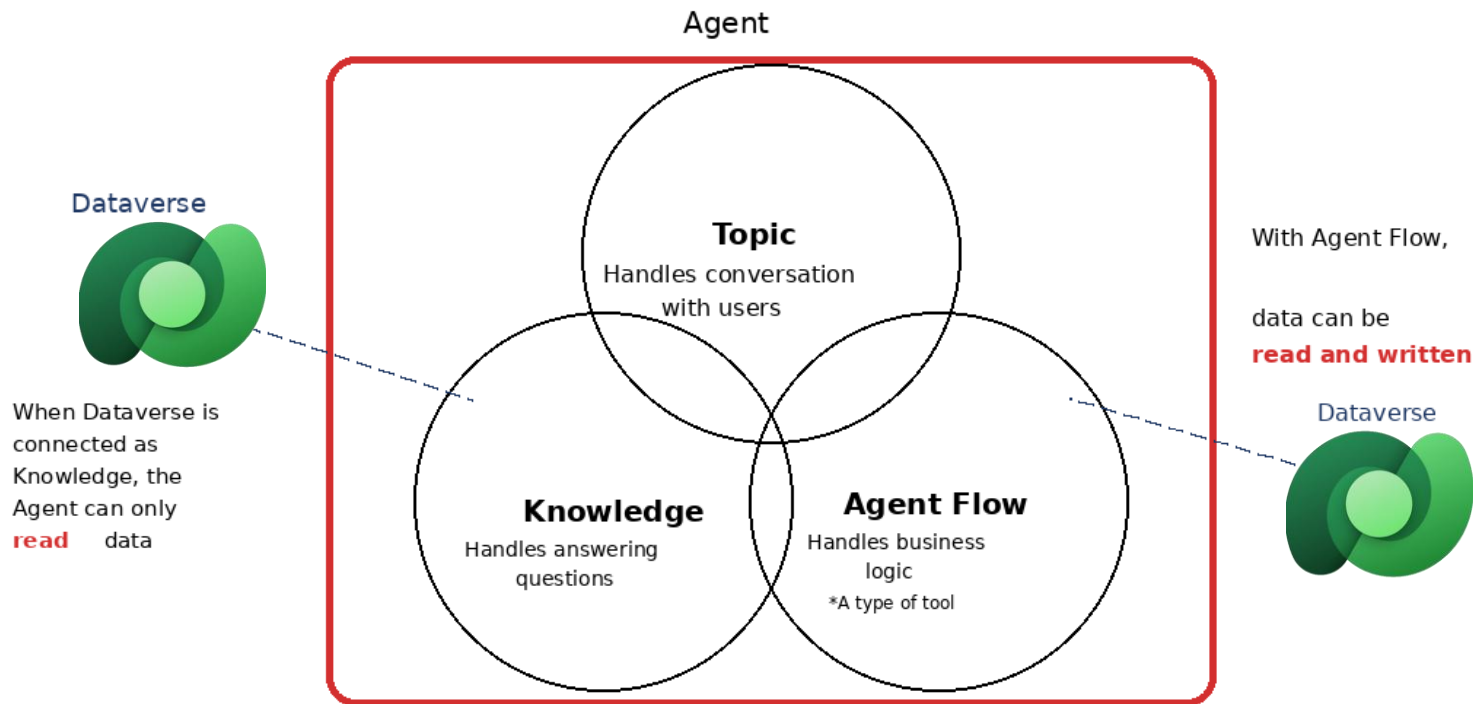
- **Automated workflow functions that can be embedded into agents**
- Think of them as reusable business logic modules that agents can call — similar to stored procedures in a database.
- Define business logic as flows with predictable, fast execution
- Power Platform connectors are available within flows
- AI actions (e.g., summarization) can be included in flows
- Defined flows can be reused across multiple agents



# Difference from **Topic**



# Difference from **Knowledge**



# Labs 8, 9 & 10: Overview

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- **Lab 8: Create Agent Flows**
  - Create "Get Property" flow (retrieve real estate info)
  - Create "Create Booking Request" flow (write booking data to Dataverse)
- **Lab 9: Configure Generative AI in Microsoft Copilot Studio**
  - Re-enable Generative AI Orchestration for production use
  - Add Dataverse source (Real Estate Property table)
- **Lab 10: Deploy Agent to Microsoft Teams**
  - Configure Entra ID authentication
  - Connect Teams & Microsoft 365 Copilot channel
  - Test the agent within Teams

# Creating an Agent Flow: Steps

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- Click "Flows" in Microsoft Copilot Studio
- Click "+ New agent flow"
- Select trigger: "When an agent calls this flow"
- Add input parameters (e.g., customer name, property name, date/time)
- Add actions (e.g., Dataverse → "Add a new row")
- Name and save the flow
- In the agent's Tools section, add the flow as an Agent Flow tool
- **The agent can now write booking information to Dataverse tables**

# Deploying to Microsoft Teams

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- Navigate to the agent's Channels settings
- Select "Teams & Microsoft 365 Copilot"
- Click "Add channel" and then "View agent in Teams"
- Open Teams (web app recommended for lab environment)
- Add and open the agent in Teams
- **The agent works identically within Teams as it does in Copilot Studio's test chat**
- Note: For customer-facing agents, channels like LINE, Web app, or native apps would be more appropriate than Teams

# Agent Publishing

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- After development and testing, publish the agent
- Publishing makes the latest agent content available on all connected channels
- Users can then interact with the agent through their preferred channel



# Overall Summary

# Summary (1/3)

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- **Microsoft Copilot Studio enables agent development with a no-code/low-code approach**
- Agents use generative AI internally
- Agents don't just answer questions — they use tools to perform tasks on behalf of users
- Autonomous agents (since October 2024) can be triggered by events like receiving an email
- Adding (Word files, websites, etc.) enables -based answers
- Agents are stored within Power Platform environments

## Summary (2/3)

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- Agents can be used through various channels: Web, Teams, Slack, LINE, Facebook, etc.
- An agent has multiple topics — each topic represents a task the agent can perform
- Topics consist of one trigger and one or more nodes
- Trigger phrases determine which topic is activated
- Agents can identify and extract entities (names, emails, dates, etc.) from user input
- Extracted entities are stored in variables for use in subsequent processing

# Summary (3/3)

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- Dataverse is Power Platform's cloud database for business data
- Dataverse for Teams is a simplified version available with Microsoft 365 licenses
- Agents can access Dataverse to retrieve and store data
- **Multiple access methods:**
  - Connect Dataverse as a source (read-only)
  - Access from topics (read/write)
  - Access from Agent Flows (read/write, reusable across agents)
- Agents can be deployed and tested within Microsoft Teams